



# CAPSTONE PROJECT: PROFESSIONAL CASE STUDY

## Hand Gesture Controller For Navigation Gameplay

| Organisation:                                   |                                                                |
|-------------------------------------------------|----------------------------------------------------------------|
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## **EXECUTIVE SUMMARY**

This report documents the development of Hand Gesture Controller for Navigation Controller, completed as a Faculty of Computer Science and Information Technology (FCSIT), Universiti Malaysia Sarawak (UNIMAS) under the supervision of my Final Year Supervisor, Dr Lim Phei Chin. The project was further aimed by participation in the K-Youth Nurture 2026 programme by Khazanah Nasional, which provided three industry simulation scenarios that significantly developed professional competencies beyond the technical scope of the FYP. The project used OpenCV, MediaPipe, MediaPipe Model Maker, WebSocket, Node.js, Phaser.js, and MySQL to create a complete gesture-to-game system. A structured literature review was conducted to identify existing approaches, followed by a user study with 31 participants to determine intuitive gestures for seven navigation actions. The system was then developed using the Rapid Application Development (RAD) methodology and evaluated through functional testing and user testing with 13 participants. The final system successfully demonstrated real-time gesture-based navigation using a laptop webcam. All 24 functional test cases were completed. All 13 participants also expressed interest in an improved version of the system. Overall, the project demonstrates that webcam-based gesture control can provide an affordable and accessible alternative to specialised gaming hardware. However, the recognition system would benefit from a larger and more diverse dataset, a larger user study, and further refinement of visually similar gestures to improve reliability for a wider user base. The project also provides a foundation that could be extended beyond gaming to areas such as education, rehabilitation, industrial human-computer interaction, and accessibility technology.

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## **1. PROJECT CONTEXT**

This capstone report is based on my Final Year Project titled “Hand Gesture Controller for Navigation Gameplay” using Computer Vision. It was completed at Faculty of Computer Science and Information Technology (FCSIT), Universiti Malaysia Sarawak (UNIMAS) under the supervision of my Final Year Supervisor, Dr Lim Phei Chin. The project was driven by a real and observable gap in the gaming industry: gesture-controlled games already existed, but they required expensive sensor motion hardware such as Xbox Kinect or Nintendo Wii Remote. Most casual users could not afford it. Simultaneously, conventional controllers such as keyboards and joysticks required repetitive button pressing that may cause finger fatigue during extended gameplay session and created an unnatural experience. The project is set out to prove that a fully functional, accessible and affordable gesture-controlled navigation game could be built using only a built-in webcam on laptop and free open-source computer vision tools.

## **2. PROBLEM IDENTIFICATION AND RE-DEFINITION**

Most people know games played with hand gesture controller are Xbox Kinect or Nintendo Wii Remote but with specific motion sensor hardware. At first, I thought gesture-controlled games are fun, intuitive and interactive, however why the lack of usability? After conducting a structured literature review of 18 research papers and existing systems of the Final Year Project: Hand Gesture Controller for Navigation Gameplay report in Chapter 2, the real problem was discovered. It was not simply that gesture games did not exist, they did. Throughout the research, it was discovered gesture controller requires an expensive extra hardware with sensor motion and some people prefer to play games without the need using the hardware due to the lack of natural interaction with the game. But, the deeper issue I discovered after implementing it, the current system required a complex local installation that non-technical users could not access completely. There was lack of affordable, hardware-independent hand gesture controller and accessibility for casual user. This re-definition changed my design approach significantly. Instead of just building the game that uses gestures, I need to build a complete production gesture-to-game system that ran on hardware that user already own with their built-in webcam, which communicated over the open using WebSocket,

without the extra installation needed. The system had to be three things simultaneously: functional, affordable and accessible.

### 3. ANALYSIS AND INSIGHTS

The analysis phase of the Final Year Project involved two distinct streams. Beforehand, the problems are framed with 5 whys.

The initial observation: Gesture controller for games can provide interactive gameplay, but existing solutions may need additional hardware, extra cost or complex setup.

| Analysis Phase       | Questions                                                                                                      | Key Insights                                                                                     |
|----------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Level 1              | What issue arise with existing gesture controller?                                                             | Need additional hardware, extra cost or complex setup                                            |
| Level 2              | Why do system need complex setup?                                                                              | Depend on specialized hardware and technical installations.                                      |
| Level 3              | Why hardware-independence important?                                                                           | Increases cost and creates accessibility barriers                                                |
| Level 4              | Why are specialized hardware unnecessary?                                                                      | Utilizing the existing laptop with webcam                                                        |
| Level 5              | Why leverage built-in webcam?                                                                                  | Computer vision software can process webcam video to track hand movements without extra hardware |
| Solution Opportunity | Develop an affordable, accessible, and functional hand gesture controller using standard, pre-existing webcams |                                                                                                  |

*Table 1: Problems framing with 5 Whys*

#### 3.1 LITERATURE REVIEW

Then it started with the first analysis, structured literature review comparing of three categories of game controllers such as sensorless controllers (keyboards, joysticks), sensor-

based controllers (Kinect, Wii, Meta Quest) and webcam-based gesture systems (OpenCV, MediaPipe, YOLOv8). A comparative table was constructed to synthesize the findings with usability, cost, accessibility and hardware dependency. The key insights from this analysis were that webcam-based gesture recognition offered the best balance of accessibility and naturalness.

### 3.2 USER STUDY ANALYSIS

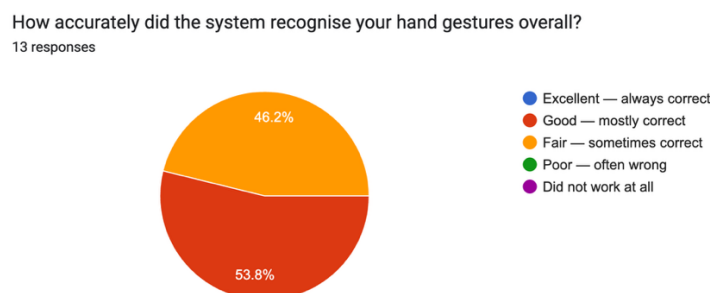
During the requirements of the project, the second analysis was a structured user study with 31 participants conducted in Phase 1 of the Rapid Application Development (RAD) methodology. Participants were asked to demonstrate their intuitive hand gesture response for seven core navigation actions after playing the game with a keyboard. Their responses were recorded and the most demonstrated gesture for each action was selected as the official gesture for the system. This finalized the gesture design in real user data rather than developer assumption. It is a critical distinction that I came to focus on user-centred design

### 3.3 TESTING ANALYSIS

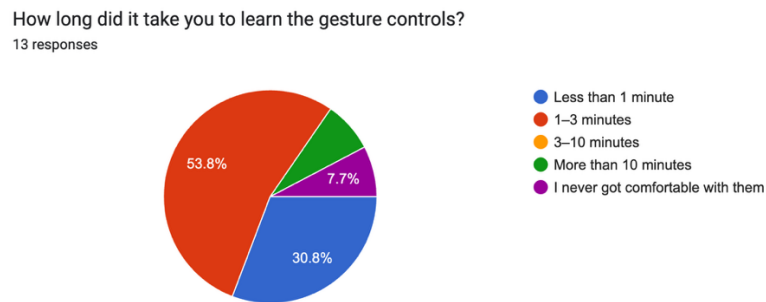
After the implementation, two rounds of system evaluation were conducted. Functional testing covered 24 test cases across user authentication, gesture recognition, game mechanics, WebSocket relay, leaderboard operations and session security. All 24 cases passed with four partial results specifically for FORWARD, BALANCE LEFT and REVERSE gestures. This is due to similarity between hand shapes caused by recognition inconsistency. Other than that, the dataset was also limited 1400 images which could not distinguish the gesture properly.

Non-functional user testing with 13 participants via Google Form in Appendix A produced quantifiable data:

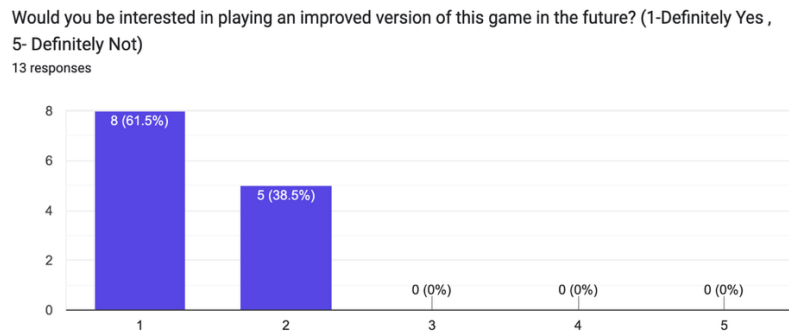
#### A. 54% rated gesture accuracy as Good



- B. 84.6% learnt the gesture controls within 3 minutes and the overall mean satisfaction rating was 4.15 out of 5.



- C. All 13 participants expressed interest in playing an improved version of the system and the game.



## 4. CREATIVE EXPLORATION

Before starting to approach the user study, several methods were considered. The first approach was to distribute Google Forms surveys accompanied by written instructions, allowing participants to complete the study independently by inserting their own video or images of the gestures. However, this method raised concerns regarding participants understanding the instructions, particularly in setting up the camera angle and performing the gestures correct which could affect the reliability of the study. Hence, the user study was selected with the combination of face-to-face sessions and online video calls with responses recorded using questionnaire in Appendix B. The face-to-face session approach helps to guide participants step by step according to the instructions. While the online sessions help to overcome the constraints of time and travelling requirements for face-to-face study along the way.

Then, to finalise system architecture, few alternative approached were evaluated after completed with literature review. The decision was to choose tool to use for gesture recognition system. Three options were considered: a custom TensorFlow model trained from scratch, YOLOv8 gesture detection and MediaPipe with Model Maker. A customer TensorFlow was rejected because it needs tens of thousands of training images and months of development. The tool was not ideal for the short term Final Year Project timeline. YOLOv8 was considered but lacked of much more detailed landmarks such as hands features. It was much more suitable for detecting standard body key points and object capabilities which do not meet the requirement needed for the Hand Gesture Controller system. Hence, the MediaPipe Model Maker was selected because it allowed training a custom gesture classifier on top of a pre-trained landmark detection model, reducing data requirements to 200 images per class while maintaining to run exceptionally fast on low-end CPUs.

## **5. PROBLEM SOLVING AND DECISION MAKING**

### **5.1 APPLE SILICON (M1) TRAINING INCOMPATIBILITY**

MediaPipe Model Maker requires tensorflow-text 2.13, which has no compatibility with Apple Silicon ARM63 wheel. Training locally on my M1 Macbook was physically impossible. Hence, the decision was made through more research from Google, AI assistance and YouTube and the model training migrate entirely to Google Colab which provides free cloud GPU access and works regardless of the developer's local hardware. Since MediaPipe is a Google owned, the workflow was recommended by Google's own MediaPipe documentation, so it was not a compromise.

### **5.2 CERTIFICATE\_VERIFY\_FAILED ON MACOS WEBSOCKET**

Throughout the implementation, there are many problems occurred. One of the main issues that I had to face because this was the most important part of the recognition system development was the failure of Python script to connect to the Render server over WSS (secure WebSocket) due to macOS Python's missing CA certificate bundle. The decision was made by adding the certify library, which provides a maintained CA certificate bundle, and pass it as the SSL context. This was a one-line fix once the root cause was identified.



### 5.3 FORWARD AND BALANCE GESTURE CONFUSION

Both gestures used a flat palm facing downward. The ML model could not distinguish them from a single frame. So, the decision making was to handle BALANCE as a post-processing rule outside the model if FORWARD is detected and the wrist has barely moved across the last 8 framed which will reclassify as BALANCE. This used temporary information that the single-frame classifier could not access.

## 6. USE OF DIGITAL TOOLS OR AI

| Tool                          | How it was used                                                                                                                                        | Decisions still made by me                                                                               |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Python + OpenCV + MediaPipe   | <ul style="list-style-type: none"><li>• Real-time webcam capture</li><li>• Hand landmark detection</li><li>• Gesture classification pipeline</li></ul> | Gesture class definitions, smoothing parameter, BALANCE post-processing rule                             |
| Google Colab                  | Training custom gesture classification model using MediaPipe Model Maker on 1400 images                                                                | Model evaluation, images updates and additions gradually, hypermeter selection (batch size per training) |
| MediaPipe Model Maker         | Custom gesture classifier training on top of pre-trained hand landmark model                                                                           | Dataset collection strategy, gesture class modelling, model export and integration                       |
| Node.js + Express + WebSocket | <ul style="list-style-type: none"><li>• Backend REST API</li><li>• JWT authentication</li><li>• WebSocket relay server</li></ul>                       | System architecture design, API endpoint structure, security implementation                              |
| Phaser.js                     | Browser-based pseudo-3D racing game engine                                                                                                             | Game mechanics design, level difficulty parameters, game penalty.                                        |

|                                 |                                                                                                                                                                                                                            |                                                                                                  |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Render + Aiven                  | Cloud deployment of game server and MySQL database management                                                                                                                                                              | Technology selection without payment, deployment configuration, database schema design           |
| GitHub                          | Version control                                                                                                                                                                                                            | Branching and commit controls, deployment activators                                             |
| Generative AI (Claude, ChatGPT) | <ul style="list-style-type: none"> <li>• Brainstorming ideas</li> <li>• Refining the existing ideas</li> <li>• Documentation sentences refinements and drafting support</li> <li>• Code review and bugs solver.</li> </ul> | All final analysis, decisions, conclusions and technical implementations were made independently |

*Table 2. Tools used throughout the project*

*AI tools were used as an assistance tool throughout this project. It was used to improve clarity of documentation and to review code structure, not to generate system logic, gestures or evaluation findings.*

## **7. OUTCOME AND IMPACT**

The system successfully completed and has achieved all three objectives:

- Studied existing systems using hand gesture controller for navigation gameplay through literature review.
- Successfully designed and implemented a hand gesture controller for navigation gameplay using computer vision tools during the implementation. The game was successfully deployed and live at <https://zenith-driven-racing-game.onrender.com> while the hand gesture recognition system can be access locally without any hardware beyond a laptop.
- The system has been evaluated the accuracy of the proposed hand gesture for navigation gameplay with user testing of 13 participants produced a mean satisfaction rating 4.15 out of 5 with the game itself has a hand gesture feedbacks after finishing the game.

| <b>Metric</b>                               | <b>Result</b>                                                                  |
|---------------------------------------------|--------------------------------------------------------------------------------|
| User study participants (gesture design)    | 31 participants, finding the most common used gestures for the 7 core gestures |
| Training images collected                   | 1400 (200 per gesture class, 7 classes)                                        |
| Functional test cases                       | 24 test case passed (4 partial)                                                |
| User testing participants                   | 13 completed the game and filled in full Google Form                           |
| Overall system satisfaction rating          | 4.15/5.0                                                                       |
| Gesture accuracy                            | 54% of participants rated Good                                                 |
| Learned gestures within 3 minutes           | 84.6% of participants, which proved the gesture naturalness                    |
| Participants interested in improved version | 100% (13/13)                                                                   |

*Table 3. Measurable outcomes produced from the projects*

The main impact of the project is improved accessibility and affordability compared with gesture-control systems that require specific hardware. This project is beneficial to casual users who do not want to spend extra cost on additional hardware on sensor motion to play a game. Users can interact with the game using hardware they already own, reducing the need to purchase, carry, or set up additional motion-sensing devices. The results also show that the gesture controls were sufficiently understandable for most participants to learn within a short period. Other than gaming, the pipeline developed from gesture dataset collection through ML training, real-time connection with WebSocket relay and cloud deployment is directly applicable to education, rehabilitation, industrial HCI and accessibility technology. However, the results also identified areas for improvement. The four partial functional results and the 54% Good accuracy rating indicate that gesture recognition remains a limitation, particularly for visually similar gestures. A larger and more diverse dataset and further gesture refinement would therefore be recommended to improve reliability for a wider range of users.

## **8. SKILLS DEMONSTRATED**

| <b>Skills</b>               | <b>Actions Demonstrated</b>                                                                                                                                                                                                                                         |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Technical Problem-Solving   | Diagnosed and resolved the Apple Silicon training incompatibility (tensorflow-text 2.13) by migrating to Google Colab.                                                                                                                                              |
| Systematically Thinking     | <ol style="list-style-type: none"> <li>1. Designed a three-layer distributed system (local Python gesture module + cloud Node.js relay + browser game) with clear separation</li> <li>2. Each layer was testable and deployable, important for debugging</li> </ol> |
| User-Centred Design         | <ol style="list-style-type: none"> <li>1. Conducted a structured user study with 31 participants before any implementation began</li> <li>2. Grounded every gesture design with user consistency</li> </ol>                                                         |
| Analytical Reasoning        | Applied the 5 whys analysis to find surface symptoms (gesture not detected) from underlying cause (low dataset diversity for geometrically similar gestures)                                                                                                        |
| Professional Communication  | <ol style="list-style-type: none"> <li>1. Interacted with 31 participants professional but welcoming, asking for their consent to participate in the user study</li> <li>2. Able to explain the instructions step by step to the participants</li> </ol>            |
| Adaptability Under Pressure | Delivered the project within the timeline and follow the amendment needed based on the supervisor and examiner instructions.                                                                                                                                        |
| Honesty in Evaluation       | Reported partial or lack of perfection test results accurately in the FYP evaluation chapter without overstating the system's performance                                                                                                                           |

*Table 4. Skills demonstrated throughout the project development*

## **9. PROFESSIONAL REFLECTION**

### **9.1 WHAT I LEARNED FROM FYP**

The Final Year Project changed how I think about building systems. Before this specific Hand Gesture Controller for Navigation Gameplay, I assumed the hardest part of any technical project would be writing the code. I finally understand that the hardest part was defining the right problem clearly enough that the code can solve. It was unsurprising, the first semester was used for defining and redefining the problem statements so it could be solved in the implementation in the next semester.

The gesture confusing between FORWARD and BALANCE is the clearest example of this. I spent significant time debugging code that was working correctly, the model was performing exactly as trained. However, the issue itself was the training data set was insufficient to create a meaningful distinction between two similar poses. The lesson I learnt, when a system fails, the first question was most likely will be directed back to the objectives and problem statement asking, “is the data right” and it is not whether the code right when the system works perfectly.

### **9.2 WHAT I WOULD DO DIFFERENTLY**

If I were to begin the FYP, I would start developing and validating with the hand gesture recognition from Phase 1 rather than Phase 3 in the RAD methodology. I would also prioritise recruiting participants earlier for a larger user study. After going through the FYP processes, I would prefer to focus on designing, refining the gestures, improving the recognition system, and finding the best approach to deploy the hand gesture controller rather than focusing on designing and developing the web game. After learning the structured way of problem solving in this programme, I would have created a clearer problem definitions and requirements in Phase 1. This would also have given time to collect much larger user study and find much more stronger consistency from the participants. Other than that, I would have much more time to collect a larger and more diverse dataset from the beginning. Although 200 images per class is sufficient for proof-of-concept, but it is sufficient to be used in production for a bigger diverse user base. Since the literature I reviewed had already identified diversity as an important

limitation in computer vision systems, I should have applied this knowledge more proactively to my own data collection strategy.

### **9.3 WHAT SKILLS IMPROVED THE MOST**

The skills I developed across the Final Year Project have given me a professional foundation that I feel equipped to build on. I can design and implement a complete fullstack system from problem definition through development. I can conduct a structure user study, analyse the results and present them in better way. I can communicate with bigger audiences, technical and non-technical in formats appropriate to each. I can stay composed when requirements change unexpectedly and adapt my output without losing quality. Most importantly, I have learnt to think before I build, to listen before I recommend and to reflect on what worked and what did not rather than presenting only the successful production. These are the habits that I believe separate a competent graduate from a genuinely professional one and they are the habits I intend to carry forward into my career.

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## APPENDICES

Appendix  
A

### HAND GESTURE FOR NAVIGATION GAMEPLAY (Final Year Project)

**NAME:** NUR FAREEHAH BINTI JOHAN  
**INSTITUTE:** UNIVERSITI MALAYSIA SARAWAK (UNIMAS)  
**COURSE:** SOFTWARE ENGINEERING, FACULTY OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY (FCSIT)  
**CONTACT NUMBER:** 0172220643  
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#### **PARTICIPANT INFORMATION AND CONSENT FORM**

Thank you for taking part in this **Final Year Project (FYP) user study**. Please read the instructions carefully and follow them accordingly. Your participation is voluntary and greatly appreciated.

#### **INSTRUCTIONS:**

1. Answer these questions: [**Circle** or **tick** your answer]

| No. | Questions                                                                            | Answers                     |       |
|-----|--------------------------------------------------------------------------------------|-----------------------------|-------|
| 1   | Which hand do you use more often? (dominant hand)                                    | left                        | right |
| 2   | Have you played games before?                                                        | yes                         | no    |
| 3   | If yes, what device do you use when playing games? (keyboard, mouse, joystick, etc.) | Please write here:<br>_____ |       |

2. You will play racing game on a laptop and **MUST** use **keyboard control only**.
  - Example of racing game: Asphalt
  - Play **at least one (1)** full game session
3. After playing the game, you will have to perform hand gestures (**your dominant hand only**) that represent the control of the gameplay instead of using keyboard controller. You will use your hand gestures that you **personally** and **naturally** feel best represent each gameplay action listed below. There are no right or wrong gestures. These are the following actions:
  - a. Start game
  - b. Accelerate/ Move forward
  - c. Reverse/ Move backward
  - d. Brake
  - e. Turn left
  - f. Turn right
  - g. Balance the car
4. A short video of your hand gesture will be recorded.
  - **Only your hand** will be recorded.

#### **CONSENT:**

|                          | <p>I, _____,</p> <p>agree to participate in this Final Year Project conducted by NUR FAREEHAH, titled:</p> <p><b>“Hand Gesture Controller for Navigation Gameplay”</b></p> <p>I understand the tasks, I agree to follow instructions, and I allow my hand gesture video to be recorded for research purposes only.</p> <p>Signature: _____</p> <p>Date: _____</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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| Appendix B               | <u><a href="#">User Testing Google Form Feedback</a></u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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| Appendix D (Gantt Chart) | <table><tr><th colspan="3">GANTT CHART</th><th colspan="3">FYP 1</th><th colspan="7">FYP 2</th></tr><tr><th colspan="3"></th><th colspan="3">2025</th><th colspan="7">2026</th></tr><tr><th>NO</th><th>DATE</th><th>AGENDA</th><th>OCT</th><th>NOV</th><th>DEC</th><th>JAN</th><th>FEB</th><th>MAR</th><th>APR</th><th>MAY</th><th>JUNE</th><th>JUL</th></tr><tr><td>1</td><td>25/10/2025</td><td>SUBMISSION OF THE APPROVED BRIEF PROPOSAL</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>2</td><td>31/10/2025</td><td>FEEDBACK &amp; COMMENT FROM EXAMINER</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>3</td><td>08/11/2025</td><td>SUBMISSION OF FULL PROPOSAL</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>4</td><td>15/11/2025</td><td>SUBMISSION OF CHAPTER 1</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>5</td><td>07/12/2025</td><td>SUBMISSION OF CHAPTER 2</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>6</td><td>28/12/2025</td><td>SUBMISSION OF CHAPTER 3</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>7</td><td>10/01/2026</td><td>SUBMISSION OF FYP1 FINAL REPORT, FYP PAPER AND VIDEO RECORDING PRESENTATION</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>8</td><td>12/01/2026</td><td>AMENDMENT &amp; 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| 1                        | 25/10/2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | SUBMISSION OF THE APPROVED BRIEF PROPOSAL                                                             |       |     |       |       |     |       |     |     |      |     |  |  |  |  |  |      |  |  |      |  |  |  |  |  |  |    |      |        |     |     |     |     |     |     |     |     |      |     |   |            |                                           |  |  |  |  |  |  |  |  |  |  |   |            |                                  |  |  |  |  |  |  |  |  |  |  |   |            |              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| 2                        | 31/10/2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | FEEDBACK & COMMENT FROM EXAMINER                                                                      |       |     |       |       |     |       |     |     |      |     |  |  |  |  |  |      |  |  |      |  |  |  |  |  |  |    |      |        |     |     |     |     |     |     |     |     |      |     |   |            |                                           |  |  |  |  |  |  |  |  |  |  |   |            |                                  |  |  |  |  |  |  |  |  |  |  |   |            |              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| 3                        | 08/11/2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | SUBMISSION OF FULL PROPOSAL                                                                           |       |     |       |       |     |       |     |     |      |     |  |  |  |  |  |      |  |  |      |  |  |  |  |  |  |    |      |        |     |     |     |     |     |     |     |     |      |     |   |            |                                           |  |  |  |  |  |  |  |  |  |  |   |            |                                  |  |  |  |  |  |  |  |  |  |  |   |            |              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| 4                        | 15/11/2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| 5                        | 07/12/2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| 6                        | 28/12/2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| 7                        | 10/01/2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | SUBMISSION OF FYP1 FINAL REPORT, FYP PAPER AND VIDEO RECORDING PRESENTATION                           |       |     |       |       |     |       |     |     |      |     |  |  |  |  |  |      |  |  |      |  |  |  |  |  |  |    |      |        |     |     |     |     |     |     |     |     |      |     |   |            |                                           |  |  |  |  |  |  |  |  |  |  |   |            |                                  |  |  |  |  |  |  |  |  |  |  |   |            |              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| 8                        | 12/01/2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| 9                        | 08/02/2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| 10                       | 09/03/2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| 11                       | 13/04/2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| 12                       | 04/05/2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | SUBMISSION OF CHAPTER 5                                                                               |       |     |       |       |     |       |     |     |      |     |  |  |  |  |  |      |  |  |      |  |  |  |  |  |  |    |      |        |     |     |     |     |     |     |     |     |      |     |   |            |                                           |  |  |  |  |  |  |  |  |  |  |   |            |                                  |  |  |  |  |  |  |  |  |  |  |   |            |              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| 13                       | 11/05/2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | SUBMISSION OF DRAFT THESIS                                                                            |       |     |       |       |     |       |     |     |      |     |  |  |  |  |  |      |  |  |      |  |  |  |  |  |  |    |      |        |     |     |     |     |     |     |     |     |      |     |   |            |                                           |  |  |  |  |  |  |  |  |  |  |   |            |                                  |  |  |  |  |  |  |  |  |  |  |   |            |              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| 14                       | 18/05/2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | SUBMISSION OF POSTER + DRAFT OF FULL PAPER                                                            |       |     |       |       |     |       |     |     |      |     |  |  |  |  |  |      |  |  |      |  |  |  |  |  |  |    |      |        |     |     |     |     |     |     |     |     |      |     |   |            |                                           |  |  |  |  |  |  |  |  |  |  |   |            |                                  |  |  |  |  |  |  |  |  |  |  |   |            |              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| 15                       | 25/05/2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | SUBMISSION OF FINAL THESIS + FULL PAPER + SOURCE CODE + INSTALLATION KITS + USER MANUAL FOR EVALUTION |       |     |       |       |     |       |     |     |      |     |  |  |  |  |  |      |  |  |      |  |  |  |  |  |  |    |      |        |     |     |     |     |     |     |     |     |      |     |   |            |                                           |  |  |  |  |  |  |  |  |  |  |   |            |                                  |  |  |  |  |  |  |  |  |  |  |   |            |              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| 16                       | 04/06/2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| 17                       | 26/06/2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | AMENDMENT PERIOD FOR FYP                                                                              |       |     |       |       |     |       |     |     |      |     |  |  |  |  |  |      |  |  |      |  |  |  |  |  |  |    |      |        |     |     |     |     |     |     |     |     |      |     |   |            |                                           |  |  |  |  |  |  |  |  |  |  |   |            |                                  |  |  |  |  |  |  |  |  |  |  |   |            |              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| 18                       | 06/07/2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | SUBMISSION OF FINAL REPORT                                                                            |       |     |       |       |     |       |     |     |      |     |  |  |  |  |  |      |  |  |      |  |  |  |  |  |  |    |      |        |     |     |     |     |     |     |     |     |      |     |   |            |                                           |  |  |  |  |  |  |  |  |  |  |   |            |                                  |  |  |  |  |  |  |  |  |  |  |   |            |              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| Appendix E               | <a href="https://zenith-driven-racing-game.onrender.com/index.html">https://zenith-driven-racing-game.onrender.com/index.html</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           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